

3.1 INTRODUCTION TO DETERMINANTS

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \det A = |A| = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = \Delta = 1 \cdot 4 - 2 \cdot 3 = -2$$

$$|A| = -2$$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \Rightarrow \det A = a_{11} \cdot a_{22} - a_{12} \cdot a_{21}$$

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow |M| = a \cdot d - b \cdot c$$

DEFINITION

For $n \geq 2$, the **determinant** of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j} \det A_{1j}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are from the first row of A . In symbols,

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + \dots + (-1)^{1+n} a_{1n} \det A_{1n}$$

$$= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{1j}$$

take any row (zeros)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\det A = (-1)^{1+1} \cdot a_{11} \cdot \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} + (-1)^{1+2} \cdot a_{12} \cdot \begin{vmatrix} 1 & 6 \\ 7 & 9 \end{vmatrix} + (-1)^{1+3} \cdot a_{13} \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} =$$

$$= 1 \cdot \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2 \cdot \begin{vmatrix} 1 & 6 \\ 7 & 9 \end{vmatrix} + 3 \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} =$$

$$= 5 \cdot 9 - 6 \cdot 8 - 2 \cdot (4 \cdot 9 - 6 \cdot 7) + 3 \cdot (4 \cdot 8 - 5 \cdot 7) = 45 - 48 - 2(36 - 42) + 3(32 - 35) = -3 - 2 \cdot (-6) + 3 \cdot (-3) = -3 + 12 - 9 = 0$$

$$M = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 4 & 0 \\ 3 & 3 & 1 \end{bmatrix} \Rightarrow \det M = (-1)^{2+1} \cdot 0 \cdot \begin{vmatrix} 2 & 5 \\ 3 & 1 \end{vmatrix} + (-1)^{2+2} \cdot 4 \cdot \begin{vmatrix} 1 & 5 \\ 3 & 1 \end{vmatrix} + (-1)^{2+3} \cdot 0 \cdot \begin{vmatrix} 1 & 2 \\ 3 & 3 \end{vmatrix} =$$

$$= 4 \cdot (1 \cdot 1 - 5 \cdot 3) = 4 \cdot (-14) = -56$$

Choose the row (column) with most zeros

3.2 PROPERTIES OF DETERMINANTS

THEOREM 3

Row Operations

Let A be a square matrix.

- If a multiple of one row of A is added to another row to produce a matrix B , then $\det B = \det A$. (Replacing does not change the $\det$)
- If two rows of A are interchanged to produce B , then $\det B = -\det A$.
- If one row of A is multiplied by k to produce B , then $\det B = k \cdot \det A$.

1) Replacing
 $R_i \leftarrow R_i + k \cdot R_j$

2) Scaling
 $k \cdot R_i$

3) Interchange
 $R_i \leftrightarrow R_j$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow \det A = 1 \cdot 4 - 2 \cdot 3 = -2$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{R_2 - 3 \cdot R_1} \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix} = B \Rightarrow \det B = 1 \cdot (-2) - 2 \cdot 0 = -2$$

A is row equiv to B

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{C_1 \leftrightarrow C_2} \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} = B \Rightarrow \det B = 2 \cdot 3 - 1 \cdot 4 = 2$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{R_1 \cdot 3} \begin{bmatrix} 3 & 6 \\ 3 & 4 \end{bmatrix} = B \Rightarrow \det B = 3 \cdot 4 - 6 \cdot 3 = -6 = 3 \cdot (-2) = \det A$$

1) Replacing does not change $\det$

2) Interchanging changes only sign of $\det$

3) Scaling scales the $\det$

4) If a matrix has a zero column or zero row then $\det = 0$

$$5) |A| = |A^T|$$

$$6) |AB| = |A| \cdot |B|$$

$$(A \ I) \rightarrow (I \ A^{-1})$$

A square matrix A is invertible if and only if $\det A \neq 0$.

EXAMPLE 3 Compute $\det A$, where $A = \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ -6 & 7 & -7 & 4 \\ -5 & -8 & 0 & 9 \end{bmatrix}$. $R_3 + 2 \cdot R_1$

SOLUTION Add 2 times row 1 to row 3 to obtain

$$\begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 5 & -3 & -6 \\ -5 & -8 & 0 & 9 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 3 & -1 & 2 & -5 \\ 0 & 5 & -3 & -6 \\ 0 & 0 & 0 & 0 \\ -5 & -8 & 0 & 9 \end{bmatrix} \rightarrow \text{cannot be reduced into } (I) \text{ because of Row 3}$$

Inverse does not exist

If A is an $n \times n$ matrix, then $\det A^T = \det A$.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$\det A =$$

$$= d \cdot a - b \cdot c$$

$$\det A^T =$$

$$= ad - cb$$

Multiplicative Property

If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$.

EXAMPLE 5 Verify Theorem 6 for $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$.

$$|A| = 6 \cdot 2 - 1 \cdot 3 = 9$$

$$|B| = 4 \cdot 2 - 1 \cdot 3 = 5$$

$$|A| \cdot |B| = 9 \cdot 5 = 45$$

$$A \cdot B = \begin{bmatrix} 6 \cdot 4 + 1 \cdot 1 & 6 \cdot 3 + 1 \cdot 2 \\ 3 \cdot 4 + 1 \cdot 2 & 3 \cdot 3 + 1 \cdot 2 \end{bmatrix} = \begin{bmatrix} 25 & 20 \\ 14 & 13 \end{bmatrix}$$

$$|AB| = 25 \cdot 13 - 20 \cdot 14 = \underline{25 \cdot 13} - \underline{20 \cdot 14} = 20 =$$

$$5 \cdot 13 - 20 = 65 - 20 = 45$$



3.3 CRAMER'S RULE, VOLUME, AND LINEAR TRANSFORMATIONS

Cramer's Rule

Cramer's rule is needed in a variety of theoretical calculations. For instance, it can be used to study how the solution of $A\mathbf{x} = \mathbf{b}$ is affected by changes in the entries of $\mathbf{b}$. However, the formula is inefficient for hand calculations, except for 2×2 or perhaps 3×3 matrices.

For any $n \times n$ matrix A and any $\mathbf{b}$ in $\mathbb{R}^n$, let $A_i(\mathbf{b})$ be the matrix obtained from A by replacing column i by the vector $\mathbf{b}$.

$$A_i(\mathbf{b}) = [\mathbf{a}_1 \quad \dots \quad \mathbf{b} \quad \dots \quad \mathbf{a}_n]$$

$\uparrow$
 col i

Cramer's Rule

Let A be an invertible $n \times n$ matrix. For any $\mathbf{b}$ in $\mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \quad i = 1, 2, \dots, n \quad (1)$$

→ What if $\det A = 0$

$$\begin{cases} x_1 + 3x_2 + x_3 = 4 \\ -x_1 + 2x_3 = 2 \\ 3x_1 + x_2 = 2 \end{cases} \quad \det A = \begin{vmatrix} 1 & 3 & 1 \\ -1 & 0 & 2 \\ 3 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 3 & 1 \\ 0 & 3 & 3 \\ 0 & -8 & -3 \end{vmatrix} = 1 \cdot \begin{vmatrix} 3 & 3 \\ -8 & -3 \end{vmatrix} = -9 + 24 = 15$$

$$\det A_1(\mathbf{b}) = \begin{vmatrix} 4 & 3 & 1 \\ 2 & 0 & 2 \\ 2 & 1 & 0 \end{vmatrix} \xrightarrow{C_1 - C_3} \begin{vmatrix} 3 & 3 & 1 \\ 0 & 0 & 2 \\ 2 & 1 & 0 \end{vmatrix} = (-1)^{2+3} \cdot 2 \cdot \begin{vmatrix} 3 & 3 \\ 2 & 1 \end{vmatrix} = -2(3 - 6) = 6$$

$$\det A_2(\mathbf{b}) = \begin{vmatrix} 1 & 4 & 1 \\ -1 & 2 & 2 \\ 3 & 2 & 0 \end{vmatrix} \xrightarrow{C_1 - C_3} \begin{vmatrix} 1 & 4 & 1 \\ 0 & 6 & 3 \\ 0 & -10 & -3 \end{vmatrix} = (-1)^{1+1} \cdot 1 \cdot \begin{vmatrix} 6 & 3 \\ -10 & -3 \end{vmatrix} = -18 + 30 = 12$$

$$\det A_3(\mathbf{b}) = \begin{vmatrix} 1 & 3 & 4 \\ -1 & 0 & 2 \\ 3 & 1 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 3 & 4 \\ 0 & 3 & 6 \\ 0 & -8 & -10 \end{vmatrix} = (-1)^{1+1} \cdot 1 \cdot \begin{vmatrix} 3 & 6 \\ -8 & -10 \end{vmatrix} = -30 + 48 = 18$$

$$x_1 = \frac{\det A_1(\mathbf{b})}{\det A} = \frac{6}{15} = \frac{2}{5}; \quad x_2 = \frac{\det A_2(\mathbf{b})}{\det A} = \frac{12}{15} = \frac{4}{5}$$

$$x_3 = \frac{\det A_3(\mathbf{b})}{\det A} = \frac{18}{15} = \frac{6}{5}; \quad \text{Ans: } \left(\frac{2}{5}; \frac{4}{5}; \frac{6}{5} \right)$$

- 1) if $\det A = 0$ and $\det A_i(\mathbf{b}) = 0$ for any i then system has inf many solutions
- 2) if $\det A = 0$ but $\exists i: \det A_i(\mathbf{b}) \neq 0$ then system has no solut.
- 3) if $\det A \neq 0$ then sys has unique solut

Application to Engineering

A number of important engineering problems, particularly in electrical engineering and control theory, can be analyzed by *Laplace transforms*. This approach converts an appropriate system of linear differential equations into a system of linear algebraic equations whose coefficients involve a parameter. The next example illustrates the type of algebraic system that may arise.

EXAMPLE 2 Consider the following system in which s is an unspecified parameter. Determine the values of s for which the system has a unique solution, and use Cramer's rule to describe the solution.

$$\begin{cases} 3sx_1 - 2x_2 = 4 \\ -6x_1 + sx_2 = 1 \end{cases}$$

- 1) if all det are zeros we have inf. many sol
- 2) if main det $A = 0$ but if at least one of others is not zero then no solut
- 3) if main det $A \neq 0$ then unique.

$$\det A = \begin{vmatrix} 3s & -2 \\ -6 & s \end{vmatrix} \neq 0$$

$$3s \cdot s - (-2) \cdot (-6) \neq 0$$

$$3s^2 - 12 \neq 0 \quad / :3$$

$$s^2 - 4 \neq 0$$

$$(s-2)(s+2) \neq 0 \Rightarrow s \neq 2 \text{ and } s \neq -2$$

An Inverse Formula

Let A be an invertible $n \times n$ matrix. Then

$$A^{-1} = \frac{1}{\det A} \text{adj } A = \frac{1}{|A|} \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

EXAMPLE 3 Find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{bmatrix}$.

$$\det A = \begin{vmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 4 & -2 \\ 2 & 1 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & -1 & 1 \\ 0 & 5 & -3 \\ 0 & 3 & 1 \end{vmatrix} =$$

$$= (-1)^{1+1} \cdot 1 \cdot \begin{vmatrix} 5 & -3 \\ 3 & 1 \end{vmatrix} = 5 \cdot 1 - (-3) \cdot 3 = 5 + 9 = 14 \neq 0 \Rightarrow A^{-1} \exists$$

$$\text{adj } A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

$$A_{11} = (-1)^{1+1} \begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3$$

$$A_{13} = (-1)^{1+3} \begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14$$

$$A_{31} = \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7$$

$$A_{32} = - \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -5$$

$$A_{33} = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$$

$$A^{-1} = \frac{1}{14} \begin{bmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -5 & -3 \end{bmatrix} = \begin{bmatrix} -1/7 & 1 & 2/7 \\ 3/14 & -1/2 & 1/14 \\ 5/14 & -5/14 & -3/14 \end{bmatrix}$$

THEOREM 9

If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$. If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$.

SG

A Geometric Proof
3-12

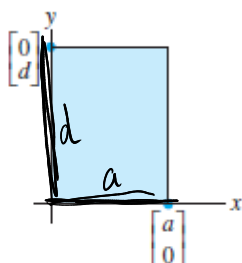


FIGURE 1

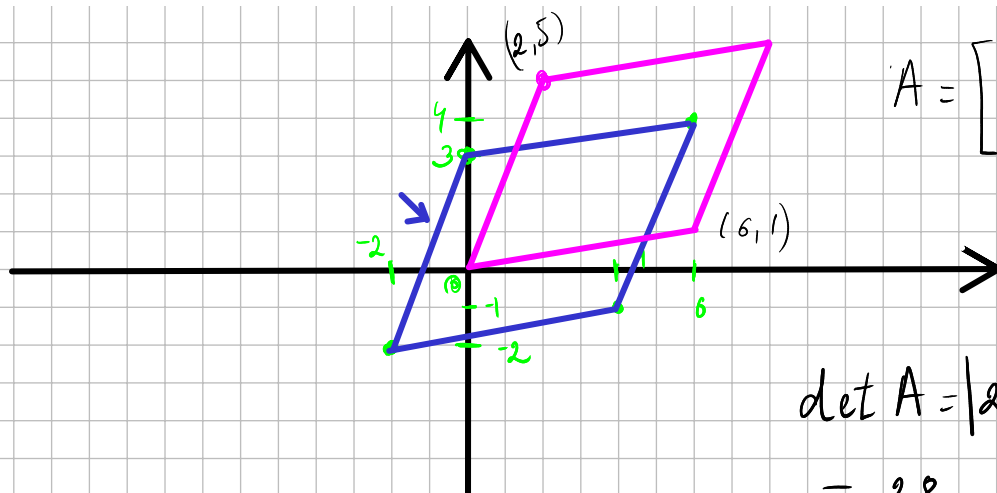
Area = $|ad|$.

PROOF The theorem is obviously true for any 2×2 diagonal matrix:

$$\left| \det \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \right| = |ad| = \begin{cases} \text{area of} \\ \text{rectangle} \end{cases}$$

See Figure 1. It will suffice to show that any 2×2 matrix $A = [\mathbf{a}_1 \ \mathbf{a}_2]$ can be transformed into a diagonal matrix in a way that changes neither the area of the associated parallelogram nor $|\det A|$. From Section 3.2, we know that the absolute value of the determinant is unchanged when two columns are interchanged or a multiple of one column is added to another. And it is easy to see that such operations suffice to transform A into a diagonal matrix. Column interchanges do not change the parallelogram at all. So it suffices to prove the following simple geometric observation that applies to vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$:

EXAMPLE 4 Calculate the area of the parallelogram determined by the points $(-2, -2)$, $(0, 3)$, $(4, -1)$, and $(6, 4)$. See Figure 5(a).



$$A = \begin{bmatrix} 2 & 6 \\ 5 & 1 \end{bmatrix}$$

$$\det A = |2 \cdot 1 - 5 \cdot 6| =$$

$$= 28$$

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear transformation determined by a 2×2 matrix A . If S is a parallelogram in $\mathbb{R}^2$, then

$$\{\text{area of } T(S)\} = |\det A| \cdot \{\text{area of } S\} \quad (5)$$

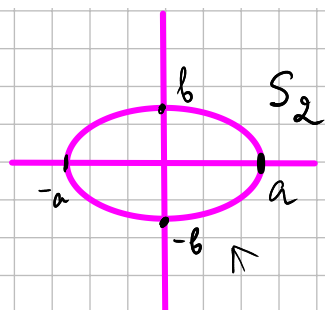
If T is determined by a 3×3 matrix A , and if S is a parallelepiped in $\mathbb{R}^3$, then

$$\{\text{volume of } T(S)\} = |\det A| \cdot \{\text{volume of } S\} \quad (6)$$

EXAMPLE 5 Let a and b be positive numbers. Find the area of the region E bounded by the ellipse whose equation is

$$\frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



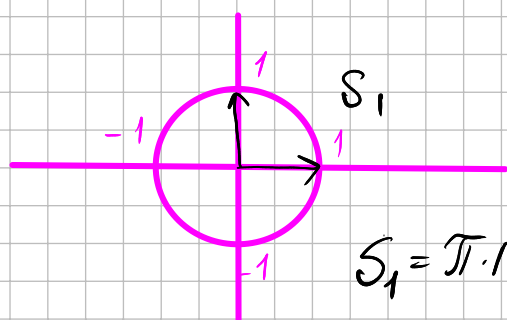
$$S = \pi \cdot a \cdot b$$

$$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$A \cdot e_1 = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ 0 \end{bmatrix} \quad \checkmark$$

$$A \cdot e_2 = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ b \end{bmatrix} \quad \checkmark$$



$$S_1 = \pi \cdot r^2 = \pi$$

$$e_1 \rightarrow \begin{bmatrix} a \\ 0 \end{bmatrix} \quad e_2 \rightarrow \begin{bmatrix} 0 \\ b \end{bmatrix}$$

$$S_2 = A \cdot S_1$$

$$\cdot \text{area } S_2 = |\det A| \cdot \text{area } S_1$$

$$\text{area } S_2 = |a \cdot b| \cdot \pi = \pi |a \cdot b| = \pi a b$$